📘 **AI/ML Internship – Day 5 Notes & Tasks**

🟢 **Module 3: Python for AI**

📌 **Part 1: Strings (Deep Understanding)**

🔹 **What is a String?**

A string is a sequence of characters used to store text data.

👉 Examples:

```
name = "AI Student"
course = 'Machine Learning'
```

🔹 **String Properties**

- Immutable (cannot change directly)
- Indexed (can access using index)
- Ordered (maintains sequence)

🔹 **Indexing**

```
text = "Python"

print(text[0])   # P
print(text[-1])  # n
```

🔹 **Slicing**

```
text = "Artificial"

print(text[0:5])   # Artif
print(text[:4])    # Arti
print(text[4:])    # ficial
```

🔹 **Important String Methods**

```
text = " hello ai "

print(text.upper())     # HELLO AI
print(text.lower())     # hello ai
print(text.strip())     # remove spaces
print(text.replace("ai", "ml"))
```

🔹 Split & Join

```
text = "AI ML Python"

words = text.split(" ")
print(words)

new_text = " ".join(words)
print(new_text)
```

🔹 Checking Data

```
text = "machine learning"

print("learn" in text)     # True
print(text.startswith("machine"))
```

📌 Part 2: Nested Data Structures (Very Important)

🔹 Why Nested Data?

In real-world AI:

- Data is complex
- Stored in structured format

🔹 List of Lists

```
matrix = [
    [1, 2],
    [3, 4]
]
```

```
print(matrix[1][0])   # 3
```

🔹 List of Dictionaries

```
students = [
    {"name": "John", "marks": 80},
    {"name": "Sara", "marks": 90}
]
```

🔹 Accessing Data

```
print(students[0]["name"])
```

🔹 Looping Nested Data

```
for student in students:
    print(student["name"], student["marks"])
```

📌 Part 3: Data Processing Logic

🔹 Example: Average Marks

```
students = [
    {"name": "A", "marks": 70},
    {"name": "B", "marks": 80}
]

total = 0

for s in students:
    total += s["marks"]

avg = total / len(students)
print(avg)
```

🔹 Example: Filter Data

```
for s in students:
    if s["marks"] > 75:
        print(s["name"])
```

📌 Part 4: Modules in Python

🔹 What is a Module?

A module is a file that contains reusable code.

🔹 Why Modules?

- Avoid code repetition
- Organized coding
- Reusability

🔹 Built-in Modules

math module

```
import math

print(math.sqrt(25))
print(math.pow(2, 3))
```

---

random module

```
import random

print(random.randint(1, 10))
```

🔹 Creating Your Own Module

👉 file: mymodule.py

```
def greet(name):
    return "Hello " + name
```

👉 main file:

```
import mymodule

print(mymodule.greet("AI Student"))
```

🎯 **Tasks for Day 5**

🧠 **Task 1: Theory**

1. What is string?
2. What is slicing?
3. What is nested data structure?
4. What is module?
5. Why modules are important?

💻 **Task 2: String Practice**

1. Reverse a string
2. Count number of words
3. Check palindrome
4. Replace words
5. Count vowels

💻 **Task 3: Nested Data Practice**

1. Create list of 5 students
2. Print all students
3. Find average marks
4. Find highest marks
5. Print students with marks > 75

💻 **Task 4: Module Practice**

1. Use math module:
    - Square root
    - Power
2. Use random module:
    - Random number
    - Random choice
3. Create your own module

🔥 **Task 6: Logic Challenge**

1. Second highest number in list
2. Count frequency of characters
3. Remove duplicates